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## Combustion and Flame

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# Experimental and kinetic modeling study of *i*-butanol pyrolysis and combustion

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## ARTICLE INFO

## Article history:

Received 16 October 2013

Received in revised form 5 February 2014

Accepted 7 February 2014

Available online xxxx

## Keywords:

*i*-Butanol

Pyrolysis

Kinetic modeling

Theoretical calculation

Unimolecular reactions

VUV photoionization mass spectrometry

## ABSTRACT

*i*-Butanol ( $\text{C}_4\text{H}_9\text{OH}$ ) pyrolysis has been studied in a flow reactor with synchrotron vacuum ultraviolet photoionization mass spectrometry combined with molecular-beam sampling technique. The pyrolysis species were identified and their mole fractions were determined. Three pressures of 30, 150 and 760 Torr were selected to study the pressure effect of *i*-butanol chemistry. A detailed kinetic model consisting of 186 species and 1294 reactions was developed to simulate *i*-butanol high temperature chemistry. To enhance the accuracy, the model was further validated by the species profiles in shock tube pyrolysis, laminar premixed flames, oxidation data from jet-stirred reactor, ignition delay times, and flame speeds. Good agreement between the predicted and measured results was obtained.

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## 1. Introduction

Synthesizing biofuels from renewable resources have the potential to relieve global environmental problems due to their neutral  $\text{CO}_2$  emission compared with that from fossil fuels. Bio-alcohols including ethanol, butanol and pentanol have attracted great attention recently, while ethanol has already been widely used as an additive to gasoline, and bio-butanol has also been tested [1]. As one of the butanol isomers, *i*-butanol can be produced from glucose by a metabolic engineering approach with high yield and specificity [2]. As a branched-chain alcohol, *i*-butanol has a higher octane number than *n*-butanol. Previous studies indicate that the heat release rate is also slightly higher and  $\text{CO}$ ,  $\text{NO}_x$  emissions, and smoke opacity are reduced by using *i*-butanol additive compared to the neat alcohol fuel [3,4].

Many previous studies have been performed on the combustion chemistry of *i*-butanol. Yasunaga et al. [5] measured concentration profiles of pyrolysis species using a gas chromatography. Stranic et al. [6] measured time histories of the species such as  $\text{OH}$ ,  $\text{H}_2\text{O}$ ,  $\text{C}_2\text{H}_4$ ,  $\text{CO}$  and  $\text{CH}_4$  using UV and IR laser absorption. Both studies were done with shock tubes. Togbé et al. [7] investigated the *i*-butanol oxidation in a jet-stirred reactor over the temperature

range of 770–1250 K at 10 atm and with equivalence ratios of 0.5–4.0. Concentration profiles of oxidation species were measured by gas chromatography and infrared spectrometry. Yang et al. [8] identified the intermediate species in rich premixed flames of four butanol isomers with the synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). Oßwald et al. [9] investigated the chemical structures of rich premixed flames of four butanol isomers ( $\phi = 1.7$ ) at 30 Torr with SVUV-PIMS and the electron-impact ionization mass spectrometry (EIMS). More recently, Hansen et al. [10] measured the flame species concentration profiles of four *i*-butanol flames at equivalence ratio of  $\phi = 1.0$ –1.5 and pressures of 15 and 30 Torr. The flame speeds and ignition delay times of *i*-butanol were also measured by various groups [5,11–13]. Based on available experimental results, several kinetic models have been proposed recently [5,12,14,15]. However, there is still room for developing an accurate detailed kinetic model of *i*-butanol combustion, especially with the emphasis on the pressure-dependent kinetics of its unimolecular decomposition sub-mechanism considering the lack of related experimental and theoretical results.

In this work, the pyrolysis of *i*-butanol in a flow reactor at pressures of 30, 150, and 760 Torr were studied to understand its high temperature chemistry. A kinetic model consisting of 186 species and 1294 reactions was developed to simulate *i*-butanol pyrolysis at various pressures. Rate constants of some important reactions

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were updated with recently calculated or experimental results. To increase the accuracy of the model, the rate constants of the  $\beta$ -scission reactions of *i*-butanol radicals ( $C_4H_8OH$ ) were calculated with the RRKM/Master Equation method. To develop the *i*-butanol mechanism over a wide range of temperature, pressure, and equivalence ratio, the species profiles in shock tube pyrolysis [6], laminar premixed flames [9,10], the oxidation data in jet-stirred reactor (JSR) [7], ignition delay times [5,11], and laminar flame speeds [16,17] were also used to validate the present model.

## 2. Experimental method

The pyrolysis of *i*-butanol in a flow tube was carried out at the National Synchrotron Radiation Laboratory in Hefei, China. Detailed descriptions of the two beamlines and pyrolysis apparatus used in this work have been published elsewhere [18–20]. The apparatus contains a pyrolysis chamber with an electronically heated furnace, a differentially pumped chamber with molecular-beam sampling system and a photoionization chamber with a home-made reflectron time-of-flight mass spectrometer (RTOF-MS). In the experiments, the mixture of Ar (970 standard cubic centimeter per minute (SCCM)) and *i*-butanol (30 SCCM in gas phase) was fed into an alumina flow tube in the pyrolysis chamber. The flow tube is 6.8-mm-ID (inner diameter) with the length of 150 mm heated by the furnace. The thermally stable  $\alpha$ -alumina used in this work only has weak catalytic effects [21]. Particularly, considering the low specific surface area inside the polished flow tube and the rapid flow velocities, the catalytic effects of alumina flow tube are negligible in this work. The pyrolysis products were sampled at 10 mm downstream the outlet by a quartz conical nozzle. The sampled pyrolysis species formed a molecular beam in the differentially pumped chamber, and passed into the photoionization chamber through a nickel skimmer. The molecular beam was crossed by the tunable VUV light in the photoionization chamber and the ionized species were detected by RTOF-MS.

The temperature distributions along the centerline of the flow tube were measured by a thermocouple before and after the pyrolysis experiment. Detailed description of the temperature measurement method has been provided in previous studies [19,22]. The measured temperature profiles in the experiments are provided as the [Supplementary Materials](#). Each profile is named by its maximum temperature ( $T_{\max}$ ) which is used as experimental temperature. The uncertainty of  $T_{\max}$  is estimated to be within  $\pm 30$  K. The pressure at the pyrolysis chamber can be changed from 5 to 760 Torr to study the pressure effect on the pyrolysis. In this work, three pressures of 30, 150, 760 Torr were chosen. Several nozzles with different orifices (70–500  $\mu\text{m}$ ) were used for experiments under different pressures.

Two experimental modes were used in this work including measurement of photoionization efficiency (PIE) spectra to identify pyrolysis products and temperature scan in order to deduce mole fraction profiles of pyrolysis species versus  $T_{\max}$ . In this work, about 30 species were detected and identified. Photon energies of 16.64, 11.70, 11.00, 10.50, 10.00 and 9.50 eV were selected to obtain near-threshold photoionization of pyrolysis species. The evaluation method of mole fraction has been reported in detail previously [18]. The photoionization cross sections (PICSSs) of *i*-butanol are taken from our previous measurement [23], which are available in our online database [24] along with the PICSSs of other pyrolysis products. The experimental uncertainties of measured mole fractions are evaluated to be  $\pm 25\%$  for products with known PICSSs and a factor of 2 for those with estimated PICSSs. The experimental C, H, and O balances are all within  $\pm 15\%$  compared with inlet fluxes of C, H, and O elements.

## 3. Theoretical methods for $\beta$ -scission reactions of $iC_4H_8OH$ radicals

Two types of reactions are considered to be important in the decomposition of *i*-butanol: unimolecular decomposition and H-abstraction followed by the  $\beta$ -scission of  $iC_4H_8OH$  radicals. The rate constants of unimolecular decomposition and  $\beta$ -scission reactions are both pressure- and temperature-dependent. Zhou et al. [25] have investigated the unimolecular dissociation reactions of *i*-butanol using high level ab initio electronic structure methods. However, they did not study the  $\beta$ -scission reactions of *i*- $C_4H_8OH$  radicals. The H-abstraction reactions of *i*-butanol lead to three *i*- $C_4H_8OH$  radicals (see [Scheme 1](#)), which subsequently decompose via  $\beta$ -scission reactions. In this work, the  $\beta$ -scission reactions of *i*- $C_4H_8OH$  radicals were theoretically investigated, and the pressure- and temperature-dependent rate constants were calculated, which are listed in [Table 1](#).

### 3.1. Potential energy profiles

Geometries and vibrational frequencies of the stationary species involved in the  $\beta$ -scission reactions of *i*- $C_4H_8OH$  radicals were calculated at the B3LYP/6-311++G(d,p) level. Single point energies were obtained using the QCISD(T) method with cc-pVTZ and cc-pVQZ basis sets of Dunning [26,27], and were extrapolated to the complete basis set limit (CBS) via the expression E1:

$$E_{\text{CBS}} = E_{\text{cc-pVQZ}} + (E_{\text{cc-pVQZ}} - E_{\text{cc-pVTZ}}) \frac{4^4}{5^4 - 4^4} \quad (\text{E1})$$

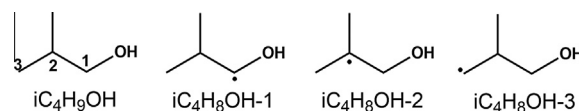
Among the reactions investigated, two dissociation reactions of the  $iC_4H_8OH$ -2 radical leading to  $iC_4H_8 + OH$  and  $iC_3H_5CH_2OH + H$ , and one dissociation reaction of  $iC_4H_8OH$ -3 radical leading to  $C_3H_6 + CH_2OH$  have no distinct transition states, so their minimum energy potentials (MEPS) were generated using the B3LYP/6-311++G(d,p) method and were scaled by the QCISD(T)/CBS limit energies. All Quantum chemical calculations were performed using the Gaussian 09 program package [28].

### 3.2. Rate constant calculations

The temperature- and pressure-dependent rate constants for the  $\beta$ -scission pathways have been computed by the RRKM/Master Equation method using the Variflex code [29]. The transition state partition functions were evaluated at the  $E/J$  (energy  $E$  and total angular momentum  $J$ ) resolved level. The step-size was chosen to be  $100 \text{ cm}^{-1}$  for  $E$  and 10 for  $J$ , respectively, which is consistent with our previous studies [30,31]. The One-dimensional (1D) master equation was solved in order to calculate pressure-dependent rate constants. The interaction between reactant and bath gas is assumed by the Lennard–Jones (L–J) model. In this work, Ar is used as the bath gas with the L–J parameters  $\sigma = 3.465 \text{ \AA}$ ,  $\epsilon/k = 113.5 \text{ K}$  [32]. For *i*- $C_4H_8OH$ , the L–J parameters are estimated by the following equations [33],

$$\sigma = 2.44(T_c/P_c)^{1/3} \quad (\text{E2})$$

$$\epsilon_A/k_b = 0.77T_c \quad (\text{E3})$$



**Scheme 1.** Schematic structures of  $iC_4H_9OH$  and three  $iC_4H_8OH$  radicals.

**Table 1**Calculation results and selected reactions in *i*-butanol model. Units are s<sup>-1</sup>, cm<sup>3</sup>, and cal/mol.

|                                                           | Selected reactions                      | <i>A</i>  | <i>n</i> | <i>E</i>  | Pressure (Torr) | Ref.              |
|-----------------------------------------------------------|-----------------------------------------|-----------|----------|-----------|-----------------|-------------------|
| <i>Reactions of iC<sub>4</sub>H<sub>9</sub>OH</i>         |                                         |           |          |           |                 |                   |
| 1                                                         | $iC_4H_9OH = iC_3H_7 + CH_2OH$          | 4.68E+119 | -30.73   | 133,860.2 | 1               | [25] <sup>a</sup> |
|                                                           |                                         | 8.65E+108 | -27.23   | 131,424.2 | 10              | [25] <sup>a</sup> |
|                                                           |                                         | 2.04E+102 | -24.97   | 131,706.3 | 100             | [25] <sup>a</sup> |
|                                                           |                                         | 5.45E+95  | -22.81   | 131,473.8 | 760             | [25] <sup>a</sup> |
|                                                           |                                         | 1.19E+83  | -18.96   | 127,108.4 | 7600            | [25] <sup>a</sup> |
|                                                           |                                         | 2.86E+64  | -13.56   | 116,986.6 | 76,000          | [25] <sup>a</sup> |
| Duplicated                                                | $iC_4H_9OH = iC_3H_7 + CH_2OH$          | 1.74E+71  | -17.07   | 101,579.4 | 1               | [25] <sup>a</sup> |
|                                                           |                                         | 4.68E+66  | -15.58   | 100,309.7 | 10              | [25] <sup>a</sup> |
|                                                           |                                         | 5.05E+61  | -13.96   | 98,980.4  | 100             | [25] <sup>a</sup> |
|                                                           |                                         | 3.21E+53  | -11.35   | 96,321.8  | 760             | [25] <sup>a</sup> |
|                                                           |                                         | 4.71E+43  | -8.27    | 92,856.5  | 7600            | [25] <sup>a</sup> |
|                                                           |                                         | 2.61E+35  | -5.71    | 89,772.7  | 76,000          | [25] <sup>a</sup> |
| 2                                                         | $iC_4H_9OH = CH_3CHCH_2OH + CH_3$       | 3.86E+123 | -32.00   | 138,756.2 | 1               | [25]              |
|                                                           |                                         | 2.93E+112 | -28.32   | 136,349.9 | 10              | [25]              |
|                                                           |                                         | 8.13E+104 | -25.73   | 136,363.8 | 100             | [25]              |
|                                                           |                                         | 7.15E+98  | -23.67   | 136,864.6 | 760             | [25]              |
|                                                           |                                         | 9.95E+24  | -2.35    | 90,670    | 7600            | [25]              |
|                                                           |                                         | 5.40E+63  | -13.26   | 120,382.4 | 76,000          | [25]              |
| Duplicated                                                | $iC_4H_9OH = CH_3CHCH_2OH + CH_3$       | 1.33E+76  | -18.54   | 107,707.3 | 1               | [25]              |
|                                                           |                                         | 2.47E+70  | -16.65   | 106,236.9 | 10              | [25]              |
|                                                           |                                         | 8.83E+65  | -15.18   | 105,173.9 | 100             | [25]              |
|                                                           |                                         | 9.67E+56  | -12.32   | 102,314.6 | 760             | [25]              |
|                                                           |                                         | 2.53E+57  | -12.32   | 102,314.6 | 7600            | [25]              |
|                                                           |                                         | 2.71E+38  | -6.52    | 95,668.1  | 76,000          | [25]              |
| 3                                                         | $iC_4H_9OH = iC_4H_8 + H_2O$            | 5.18E+94  | -23.93   | 11,2376.8 | 1               | [25] <sup>b</sup> |
|                                                           |                                         | 4.39E+89  | -22.19   | 11,2716.5 | 10              | [25] <sup>b</sup> |
|                                                           |                                         | 4.73E+82  | -19.93   | 111,907.8 | 100             | [25] <sup>b</sup> |
|                                                           |                                         | 8.57E+71  | -16.66   | 107,794.8 | 760             | [25] <sup>b</sup> |
|                                                           |                                         | 7.51E+51  | -10.82   | 96,534.4  | 7600            | [25] <sup>b</sup> |
|                                                           |                                         | 1.02E+28  | -4.00    | 80,777.5  | 76,000          | [25] <sup>b</sup> |
| Duplicated                                                | $iC_4H_9OH = iC_4H_8 + H_2O$            | 1.62E+42  | -9.46    | 74,756.9  | 1               | [25] <sup>b</sup> |
|                                                           |                                         | 1.13E+39  | -8.40    | 74,154.8  | 10              | [25] <sup>b</sup> |
|                                                           |                                         | 1.38E+32  | -6.17    | 72,092.3  | 100             | [25] <sup>b</sup> |
|                                                           |                                         | 1.24E+25  | -3.95    | 69,733.8  | 760             | [25] <sup>b</sup> |
|                                                           |                                         | 1.22E+17  | -1.46    | 66,717.5  | 7600            | [25] <sup>b</sup> |
|                                                           |                                         | 1.40E+10  | 0.61     | 63,605.9  | 76,000          | [25] <sup>b</sup> |
| 4                                                         | $iC_4H_9OH + H = iC_4H_8OH-1 + H_2$     | 8.80E+04  | 2.70     | 2915      |                 | [14]              |
| 5                                                         | $iC_4H_9OH + H = iC_4H_8OH-2 + H_2$     | 1.20E+06  | 2.40     | 4471      |                 | [6]               |
| 6                                                         | $iC_4H_9OH + H = iC_4H_8OH-3 + H_2$     | 6.66E+05  | 2.50     | 6756      |                 | [6]               |
| 7                                                         | $iC_4H_9OH + H = iC_4H_9O + H_2$        | 4.72E+02  | 3.10     | 8701      |                 | [6]               |
| 8                                                         | $iC_4H_9OH + OH = iC_4H_8OH-1 + H_2O$   | 3.61E+03  | 2.89     | -2331     |                 | [14]              |
| 9                                                         | $iC_4H_9OH + OH = iC_4H_8OH-2 + H_2O$   | 1.54E+00  | 3.70     | -4940     |                 | [14]              |
| 10                                                        | $iC_4H_9OH + OH = iC_4H_8OH-3 + H_2O$   | 5.40E+06  | 2.00     | 512       |                 | [49]              |
| 11                                                        | $iC_4H_9OH + OH = iC_4H_9O + H_2O$      | 5.88E+02  | 2.82     | -585      |                 | [14]              |
| 12                                                        | $iC_4H_9OH + CH_3 = iC_4H_8OH-1 + CH_4$ | 1.53E+01  | 3.31     | 6950      |                 | [46]              |
| 13                                                        | $iC_4H_9OH + CH_3 = iC_4H_8OH-2 + CH_4$ | 7.26E+01  | 3.13     | 6290      |                 | [46]              |
| 14                                                        | $iC_4H_9OH + CH_3 = iC_4H_8OH-3 + CH_4$ | 1.55E+01  | 3.41     | 9950      |                 | [46]              |
| 15                                                        | $iC_4H_9OH + CH_3 = iC_4H_9O + CH_4$    | 1.44E+01  | 3.1      | 6935      |                 | [46]              |
| <i>Reactions of iC<sub>4</sub>H<sub>8</sub>OH isomers</i> |                                         |           |          |           |                 |                   |
| 16                                                        | $iC_4H_8OH-1 = CH_3CHCHOH + CH_3$       | 3.37E+18  | -3.05    | 20,907    | 5               | This work         |
|                                                           |                                         | 2.94E+22  | -3.93    | 24,108    | 30              | This work         |
|                                                           |                                         | 3.67E+24  | -4.41    | 25,895    | 70              | This work         |
|                                                           |                                         | 4.23E+27  | -5.13    | 28,536    | 200             | This work         |
|                                                           |                                         | 7.61E+31  | -6.15    | 32,349    | 760             | This work         |
|                                                           |                                         | 7.70E+39  | -8.08    | 40,074    | 7600            | This work         |
|                                                           |                                         | 6.44E+45  | -9.41    | 47,205    | 76,000          | This work         |
| 17                                                        | $iC_4H_8OH-1 = iC_3H_7CHO + H$          | 1.19E+17  | -2.915   | 21,060.9  | 5               | This work         |
|                                                           |                                         | 2.59E+21  | -3.893   | 24,472.5  | 30              | This work         |
|                                                           |                                         | 4.64E+23  | -4.411   | 26,343.3  | 70              | This work         |
|                                                           |                                         | 5.68E+26  | -5.134   | 28,987.8  | 200             | This work         |
|                                                           |                                         | 1.57E+31  | -6.195   | 32,924.7  | 760             | This work         |
|                                                           |                                         | 2.88E+39  | -8.180   | 40,864.0  | 7600            | This work         |
|                                                           |                                         | 6.50E+45  | -9.616   | 48,377.6  | 76,000          | This work         |
| 18                                                        | $iC_4H_8OH-1 = iC_3H_6CHOH + H$         | 7.46E+14  | -2.644   | 21,322    | 5               | This work         |
|                                                           |                                         | 3.40E+19  | -3.693   | 24,808    | 30              | This work         |
|                                                           |                                         | 8.85E+21  | -4.248   | 26,742    | 70              | This work         |
|                                                           |                                         | 2.07E+25  | -5.040   | 29,538    | 200             | This work         |
|                                                           |                                         | 9.08E+29  | -6.142   | 33,602    | 760             | This work         |
|                                                           |                                         | 6.50E+38  | -8.270   | 41,968    | 7600            | This work         |
|                                                           |                                         | 6.25E+45  | -9.857   | 50,051    | 76,000          | This work         |

(continued on next page)

Table 1 (continued)

|            | Selected reactions                                                                        | A        | n      | E        | Pressure (Torr) | Ref.      |
|------------|-------------------------------------------------------------------------------------------|----------|--------|----------|-----------------|-----------|
| 19         | $i\text{C}_4\text{H}_8\text{OH}-2 = i\text{C}_4\text{H}_8 + \text{OH}$                    | 1.02E+08 | −0.253 | 9250     | 5               | This work |
|            |                                                                                           | 1.86E+10 | −0.681 | 10,869   | 30              | This work |
|            |                                                                                           | 3.77E+11 | −0.949 | 11,875   | 70              | This work |
|            |                                                                                           | 2.80E+13 | −1.352 | 13,386   | 200             | This work |
|            |                                                                                           | 1.63E+16 | −1.970 | 15,757   | 760             | This work |
|            |                                                                                           | 2.56E+22 | −3.429 | 21,442   | 7600            | This work |
|            |                                                                                           | 2.18E+28 | −4.805 | 27,870   | 76,000          | This work |
| 20         | $i\text{C}_4\text{H}_8\text{OH}-2 = i\text{C}_3\text{H}_6\text{CHOH} + \text{H}$          | 1.13E−09 | 3.483  | 11,412.5 | 5               | This work |
|            |                                                                                           | 1.63E−01 | 1.528  | 13,592.6 | 30              | This work |
|            |                                                                                           | 3.97E+02 | 0.751  | 15,021.3 | 70              | This work |
|            |                                                                                           | 4.05E+06 | −0.157 | 17,118.7 | 200             | This work |
|            |                                                                                           | 4.75E+11 | −1.303 | 20,383.5 | 760             | This work |
|            |                                                                                           | 4.33E+21 | −3.627 | 28,298.1 | 7600            | This work |
|            |                                                                                           | 1.55E+32 | −6.118 | 38,535.4 | 76,000          | This work |
| 21         | $i\text{C}_4\text{H}_8\text{OH}-2 = i\text{C}_3\text{H}_5\text{CH}_2\text{OH} + \text{H}$ | 2.33E−16 | 5.070  | 12,056   | 5               | This work |
|            |                                                                                           | 1.98E−07 | 2.879  | 12,833   | 30              | This work |
|            |                                                                                           | 1.47E−03 | 1.970  | 13,984   | 70              | This work |
|            |                                                                                           | 3.76E+01 | 0.963  | 15,851   | 200             | This work |
|            |                                                                                           | 5.47E+06 | −0.190 | 18,752   | 760             | This work |
|            |                                                                                           | 3.45E+16 | −2.433 | 26,200   | 7600            | This work |
|            |                                                                                           | 1.83E+28 | −5.218 | 37,262   | 76,000          | This work |
| Duplicated | $i\text{C}_4\text{H}_8\text{OH}-2 = i\text{C}_3\text{H}_5\text{CH}_2\text{OH} + \text{H}$ | 1.29E−16 | 5.196  | 10,637   | 5               | This work |
|            |                                                                                           | 2.88E−08 | 3.170  | 11,327   | 30              | This work |
|            |                                                                                           | 2.02E−04 | 2.268  | 12,588   | 70              | This work |
|            |                                                                                           | 1.34E+00 | 1.421  | 14,112   | 200             | This work |
|            |                                                                                           | 8.76E+04 | 0.361  | 16,827   | 760             | This work |
|            |                                                                                           | 3.44E+13 | −1.556 | 23,371   | 7600            | This work |
|            |                                                                                           | 3.32E+35 | −7.193 | 42,652   | 76,000          | This work |
| 22         | $i\text{C}_4\text{H}_8\text{OH}-3 = \text{C}_3\text{H}_6 + \text{CH}_2\text{OH}$          | 1.29E+12 | −1.241 | 14,224   | 5               | This work |
|            |                                                                                           | 1.69E+15 | −1.904 | 16,616   | 30              | This work |
|            |                                                                                           | 1.09E+17 | −2.312 | 18,066   | 70              | This work |
|            |                                                                                           | 3.58E+19 | −2.893 | 20,150   | 200             | This work |
|            |                                                                                           | 1.67E+23 | −3.762 | 23,321   | 760             | This work |
|            |                                                                                           | 4.82E+30 | −5.570 | 30,222   | 7600            | This work |
|            |                                                                                           | 4.90E+36 | −6.962 | 36,963   | 76,000          | This work |
| 23         | $i\text{C}_4\text{H}_8\text{OH}-3 = \text{C}_3\text{H}_5\text{OH} + \text{CH}_3$          | 3.52E+06 | −0.002 | 14,382   | 5               | This work |
|            |                                                                                           | 3.93E+11 | −1.153 | 17,480   | 30              | This work |
|            |                                                                                           | 1.20E+14 | −1.724 | 19,207   | 70              | This work |
|            |                                                                                           | 2.61E+17 | −2.502 | 21,695   | 200             | This work |
|            |                                                                                           | 8.48E+21 | −3.566 | 25,327   | 760             | This work |
|            |                                                                                           | 1.39E+31 | −5.798 | 33,456   | 7600            | This work |
|            |                                                                                           | 1.32E+39 | −7.675 | 41,894   | 76,000          | This work |
| 24         | $i\text{C}_4\text{H}_8\text{OH}-3 = i\text{C}_3\text{H}_5\text{CH}_2\text{OH} + \text{H}$ | 1.51E−04 | 2.204  | 13,562   | 5               | This work |
|            |                                                                                           | 3.44E+04 | 0.141  | 17,593   | 30              | This work |
|            |                                                                                           | 1.32E+08 | −0.719 | 19,678   | 70              | This work |
|            |                                                                                           | 2.71E+12 | −1.744 | 22,495   | 200             | This work |
|            |                                                                                           | 1.37E+18 | −3.104 | 26,692   | 760             | This work |
|            |                                                                                           | 9.04E+28 | −5.716 | 35,832   | 7600            | This work |
|            |                                                                                           | 6.43E+38 | −8.044 | 45,871   | 76,000          | This work |

<sup>a</sup> The reaction rates are divided by a factor of 2 to accord with that measured by Rosado-Reyes and Tsang [42].

<sup>b</sup> The reaction rates are multiplied by a factor of 2.4 to accord with that measured by Rosado-Reyes et al. [43].

where  $k_b$  is the Boltzmann constant,  $T_c$  and  $P_c$  are the critical temperature and pressure, respectively. We determined the value of  $T_c$  and  $P_c$  using the method of Joback in the book of Poling et al. [34]. Thereby the L–J parameters for  $i\text{-C}_4\text{H}_8\text{OH}$  radicals are defined as  $\sigma = 5.511 \text{ \AA}$ ,  $\epsilon/k = 422.01 \text{ K}$ . An exponential-down model with  $\langle \Delta E_{\text{down}} \rangle = 150 \times (T/300)^{0.85} \text{ cm}^{-1}$  was used to treat the collision energy transfer probabilities. Reactions with apparent transition states were treated with the conventional transition state theory [35], in which the rigid-rotor harmonic-oscillator (RRHO) assumption was applied. And the rate constants were treated with the Eckart tunneling corrections [36]. For barrierless reactions, rate constants were calculated using a variational selection method of transition states following the procedure proposed by da Silva and Bozzelli [37]. In brief, the locations of the optimal transition state varied with temperature. The rate constants were calculated as a

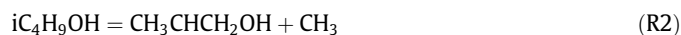
function of temperature and position along the MEPs, and then minimized to get the minimal rate constants.

#### 4. Kinetic model

In this work, the simulations of pyrolysis, oxidation, premixed flame, ignition delay times and flame speeds were performed using the Chemkin-Pro software [38]. Detailed simulation methods are described in the relevant sections. The present model consisting of 186 species and 1294 reactions was mainly developed from our reported  $n$ -butanol [22] and 2-butanol models [39], which were built based on the USC Mech II model [40] and the three butene isomers model [19] by our group. Some selected reactions in the  $i$ -butanol sub-mechanism are listed in Table 1. Most thermodynamic and transport data were taken from the USC Mech II model

[40], the database reported by Goos, Burcat and Ruscic [41], and previous butanol models [12,14]. The reaction mechanism, thermodynamic and transport data are available in the [Supplementary Materials](#).

The thermal decomposition mechanism of *i*-butanol was recently investigated by Zhou et al. [25] at the CCSD(T)/CBS//MP2/6-311G(d,p) level of theory. The pressure-dependent rate coefficients of *i*-butanol unimolecular decomposition reactions (R1–R3) were calculated in their work [25], and the sum of two modified Arrhenius expressions were used to accurately fit the rate constants in the temperature range of 500–2000 K. Recently, Rosado-Reyes and Tsang [42] measured the rate coefficients of *i*-butanol C–C scission reactions (R1 and R2) in a shock tube at temperatures of 1090 to 1240 K and pressures of 1.5 and 6 atm. As shown in Fig. 1, the rate coefficients of R1 measured by Rosado-Reyes and Tsang [42] are around half of those calculated by Zhou et al. [25]. Moreover, the calculated  $k_2$  is slightly under-estimated by a factor of 1.5 when compared to the measured results. In this work, the pressure-dependent rate coefficients of R1 and R2 from Zhou et al. [25] were used, but the rate coefficient of R1 was divided by a factor of 2 to agree with the measured values of Rosado-Reyes and Tsang [42].



Rosado-Reyes et al. [43] also measured the dehydration rate coefficients of *i*-butanol (R3) in single pulse shock tube at 1.5–6 atm and 1090–1240 K using GC/MS. They also calculated the rate coefficients of R3 via multi-structural transition state theory (MSTST). It is found that the rate coefficient of R3 calculated by Rosado-Reyes et al. agrees well with that calculated by Zhou et al. [25], but underestimated by approximately a factor of 2 compared to the shock tube measurement. The rate coefficients of R3 used in this

work were taken from Zhou et al. [25], but multiplied by a factor of 2.4 to agree with that measured values by Rosado-Reyes et al. [43].

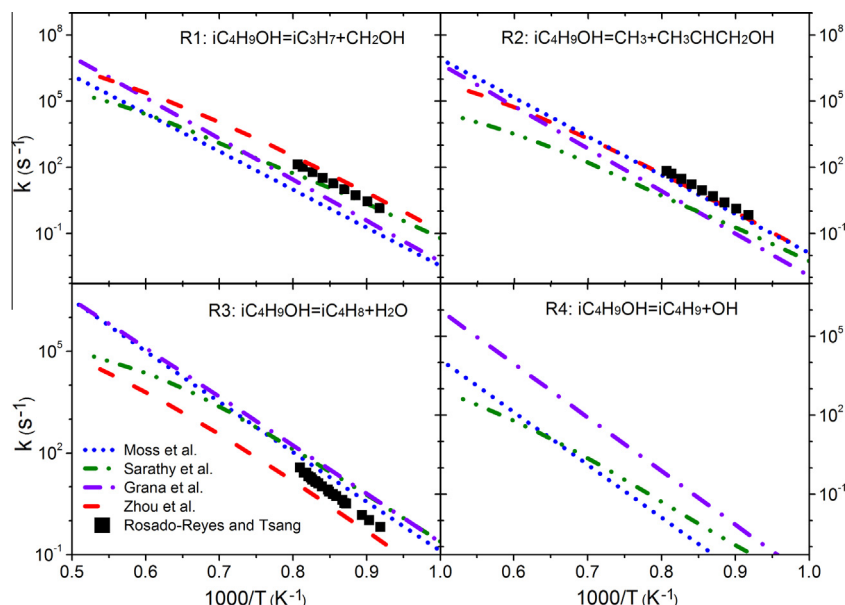
Rate constants of  $\text{iC}_4\text{H}_9\text{OH} + \text{H}$  reactions have not been studied in detail yet. In previous models, most rate constants for the  $\text{iC}_4\text{H}_9\text{OH} + \text{H}$  reactions were estimated with analogous abstraction reactions such as ethanol + H or alkane + H. Therefore, the uncertainty of these rates are at least a factor of 2 [6]. Rate constants of the H-atom abstraction reaction in this work were taken from the estimation by Sarathy et al. [14] and Stranic et al. [6]. It is noteworthy that the simulated results of *i*-butanol shock tube pyrolysis experiment [6] are very sensitive to the  $\text{iC}_4\text{H}_9\text{OH} + \text{H}$  reactions, these will be discussed in Section 5.2.

Recently, Pang et al. [44] measured overall rate constants of the  $\text{iC}_4\text{H}_9\text{OH} + \text{OH}$  reactions in shock-heated mixtures of tert-butylhydroperoxide (TBHP) with *i*-butanol largely diluted in argon. They also compared their measured rate constants with those used in previous models [14,15]. It was found that the rate constants used in Sarathy et al. model [14] and Grana et al. model [15] are slower than the measurements, while the ones used by Hansen et al. [10] show the best agreement with the experiment. Thus the  $\text{iC}_4\text{H}_9\text{OH} + \text{OH}$  rate constants from Hansen et al. [10] were used in the present model.

## 5. Results and discussion

### 5.1. *i*-Butanol pyrolysis in flow reactor

The pyrolysis experiments were simulated using the Plug Flow Code within the Chemkin-Pro program. The measured temperature profiles were used as the input temperatures. The residence times were calculated to be about 5 ms at 30 Torr and 100 ms at 760 Torr with Chemkin program, respectively, based on the input temperature profiles, volumetric flow rate (1000 SCCM) and the diameter of the flow tube (0.68 cm). The length of the flow reactor is 208 mm, mole fractions of *i*-butanol and pyrolysis products were calculated along the flow tube. Since the species were sampled at 10 mm downstream the outlet of the flow tube (at position 208 mm in the temperature profiles), only the simulated results at this position were compared with the experimental data.



**Fig. 1.** Reaction rate constants of R1–R4 at 760 Torr in Moss et al. model [12], Sarathy et al. model [14], Grana et al. model [15], calculated results by Zhou et al. [25] and measurement by Rosado-Reyes et al. [42,43].



Generally, the present model predicts most of the detected species within the experimental uncertainty. We will discuss the results in sequence of major products, oxygenated species, and small hydrocarbons below.

Figure 2 shows the decomposition profiles of *i*-butanol as well as the primary products  $H_2$ ,  $H_2O$  and  $iC_4H_8$ . The rate of production (ROP) analysis was performed to analyze *i*-butanol chemistry at 30 Torr, 1300 K and 760 Torr, 1100 K, where most of the species reach their peak values. The unimolecular decomposition of *i*-butanol is the initial step in *i*-butanol pyrolysis, while the ROP analysis shows that 25% of *i*-butanol is consumed by the unimolecular reactions (R1–R3) at 30 Torr, decreasing to 13% at 760 Torr. Most of *i*-butanol decomposes through H-abstraction reactions by H, OH and  $CH_3$  attack. The C1–C2 (R1) bond is the weakest bond in *i*-butanol; approximately 18% of *i*-butanol decomposes through the C1–C2 bond scission, producing  $iC_3H_7$  and  $CH_2OH$  radicals. The contribution of C2–C3 scission (R2, 3%) is much smaller than C1–C2 scission due to the higher bond dissociation energy. As shown in Fig. 3, R1 has the greatest sensitivity to the *i*-butanol decomposition, not only because R1 is dominating among the unimolecular reactions of *i*-butanol, but also because further destruction of  $iC_3H_7$  (R5) and  $CH_2OH$  (R6) radicals leads to the formation of  $C_3H_6$ ,  $CH_2O$  and large amount of H atoms, while H atom then attacks *i*-butanol and consequently accelerates *i*-butanol decomposition. Figure 4 illustrates the comparison of experimental and simulated results of *i*-butanol decomposition at three pressures using different sources for rate constants of R1. It can be seen that the rate constants measured by Rosado Reyes and Tsang [42] provide a better prediction for the decomposition of *i*-butanol, while those calculated by Zhou et al. over-predict the reactivity of the pyrolysis of *i*-butanol at all three pressures.[42]. It is noteworthy that the *i*-butanol pyrolysis in a shock tube performed by Stranic et al. [6] also shows that the calculated  $k_1$  by Zhou et al. is slightly overestimated, which will be discussed in detail in the next section.

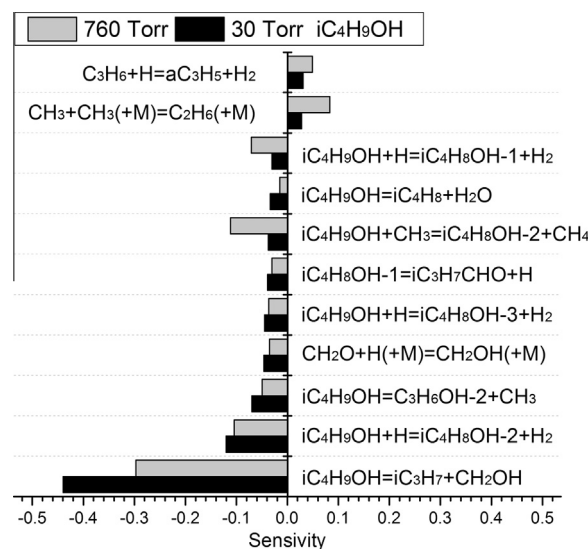


Fig. 3. Sensitivity analysis of  $iC_4H_9OH$  pyrolysis in laminar flow reactor at  $T_{max} = 1300$  K under 30 Torr (black) and  $T_{max} = 1100$  K under 760 Torr (gray). Only those reactions with sensitivities greater than 0.02 are listed.

ROP analysis and sensitivity analysis show that unimolecular reactions at 30 Torr are more dominant than those at 760 Torr. So, the unimolecular pathways are mainly discussed in the context of 30 Torr in this section. About 6% of *i*-butanol decomposes through  $H_2O$  elimination reactions producing  $iC_4H_8$  (R3). This ratio is much smaller than that in *n*-butanol [22] and 2-butanol pyrolysis [39], mainly because there is only one  $\beta$ -H in *i*-butanol. The measured and simulated mole fractions of  $iC_4H_8$  and  $H_2O$  are illustrated in Fig. 2. A ROP analysis shows that only 19% of  $iC_4H_8$  is produced by the  $H_2O$  elimination reaction (R3) at 30 Torr and this ratio decreases to 7% at 760 Torr. Most of  $iC_4H_8$  is produced through the reaction sequence of  $iC_4H_9OH + H/OH/CH_3 \rightarrow iC_4H_8OH-2 \rightarrow iC_4H_8$ . The sensitivity analysis of  $iC_4H_8$  presented in Fig. 5 shows that

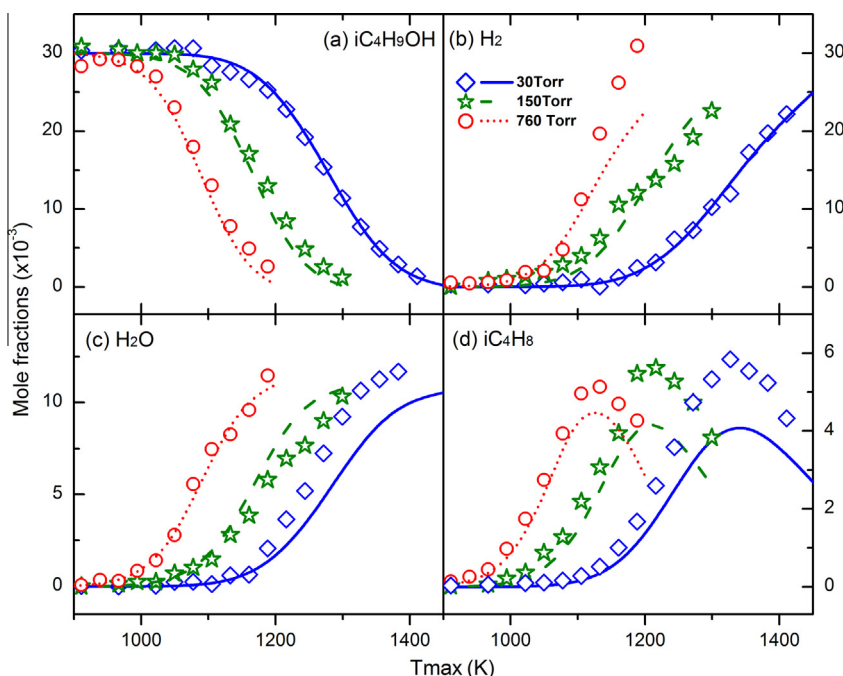
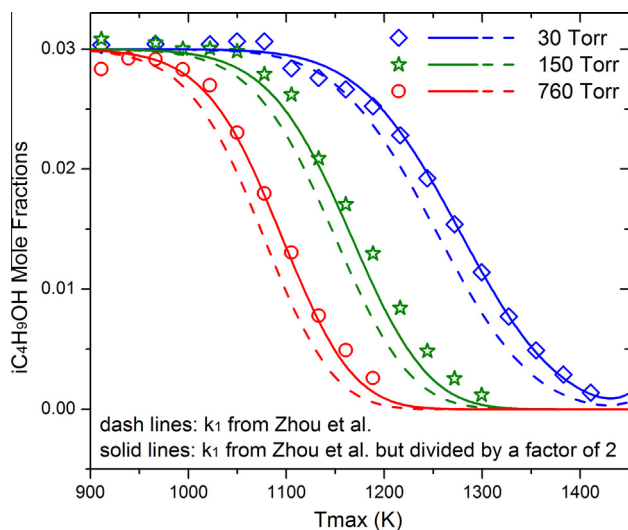


Fig. 2. Experimental mole fraction profiles (symbols) and modeling results (lines) of  $iC_4H_9OH$ ,  $H_2$ ,  $H_2O$ , and  $iC_4H_8$  from *i*-butanol pyrolysis in a laminar flow reactor at various pressures.

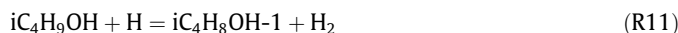
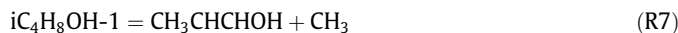


**Fig. 4.** Experimental mole fraction profiles (symbols) and modeling results (lines) of  $iC_4H_9OH$  pyrolysis using different  $k_1$ . Dash lines:  $k_1$  from Zhou et al.'s calculation [25]; solid lines:  $k_1$  from Zhou et al. but divided by a factor of 2 to agree with that measured by Rosado-Reyes and Tsang [42].

the H-abstraction reactions of  $i$ -butanol producing  $iC_4H_8OH-2$  are extremely important at both low and high pressures to  $iC_4H_8$  formation. Most of  $iC_4H_8$  decomposes via further H-abstraction reactions to form  $iC_4H_7$  radical.

$iC_4H_8OH-1$  is a key product of H-abstraction reactions from  $i$ -butanol. The further destruction of  $iC_4H_8OH-1$  leads to the formation of four C3 and C4 oxygenated species, which are illustrated in Fig. 6. More than 90% of  $iC_4H_8OH-1$  decomposes to  $CH_3CHCHOH$  (propenol) through  $\beta$ -C-C scission (R7), which is also the dominant pathway of  $CH_3CHCHOH$  formation. Most of  $CH_3CHCHOH$  is consumed through the tautomerization reaction producing more stable species  $C_2H_5CHO$ . As shown in Fig. 6, the mole fractions of  $CH_3CHCHOH$  and  $C_2H_5CHO$  are under-predicted by the present model. The sensitivity analysis shows that the H atom abstraction reactions of  $i$ -butanol producing  $iC_4H_8OH-1$  (R11) as well as the isomerization reaction of  $CH_3CHCHOH$  (R9) are the top 2 sensitive reactions to  $CH_3CHCHOH$  and  $C_2H_5CHO$  formation. The

rate constants of  $CH_3CHCHOH$  isomerization reactions (R9, R10) are estimated by analogy to  $C_2H_5OH$  (ethanol), and further calculated or experimental results for these reactions are desired. It is noteworthy that the  $CH_3CHCHOH$  photoionization cross section was estimated in this work, which causes the uncertainty of the  $CH_3CHCHOH$  mole fraction around a factor of 2. Moreover, subtraction of the  $CH_3CHCHOH$  signal from  $C_2H_5CHO$  signal is necessary when calculating the  $C_2H_5CHO$  mole fraction, thus the uncertainty of  $CH_3CHCHOH$  photoionization cross section will also lead to large uncertainty of  $C_2H_5CHO$  mole fraction.

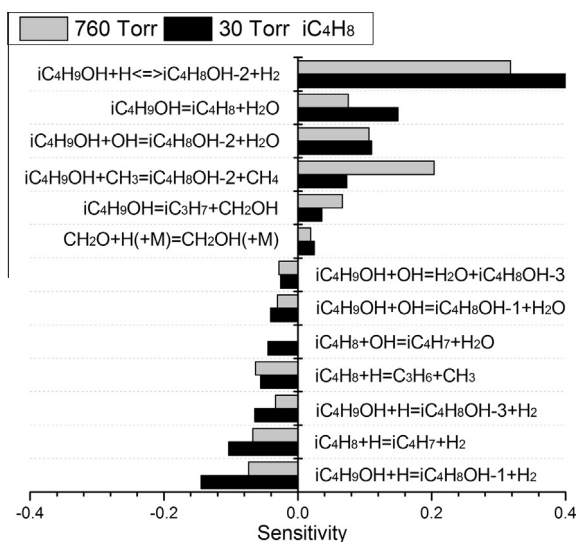


About 10% of  $iC_4H_8OH-1$  decomposes through  $\beta$ -C-H scission, producing  $iC_3H_7CHO$ . The C-C scission reactions and H-abstraction reactions of  $iC_3H_7CHO$  are included in the present model, whose rate constants are estimated by analogy to  $CH_3CHO$  [40].  $iC_3H_7CHO$  was detected in the experiment, however, its peak mole fraction is about 20 times lower than that of  $CH_3CHCHOH$ .  $iC_3H_7CHO$  is slightly over-predicted by the present model (see Fig. 6), especially at high pressures, indicating that the sub-mechanism of  $iC_3H_7CHO$  needs further development and validation.

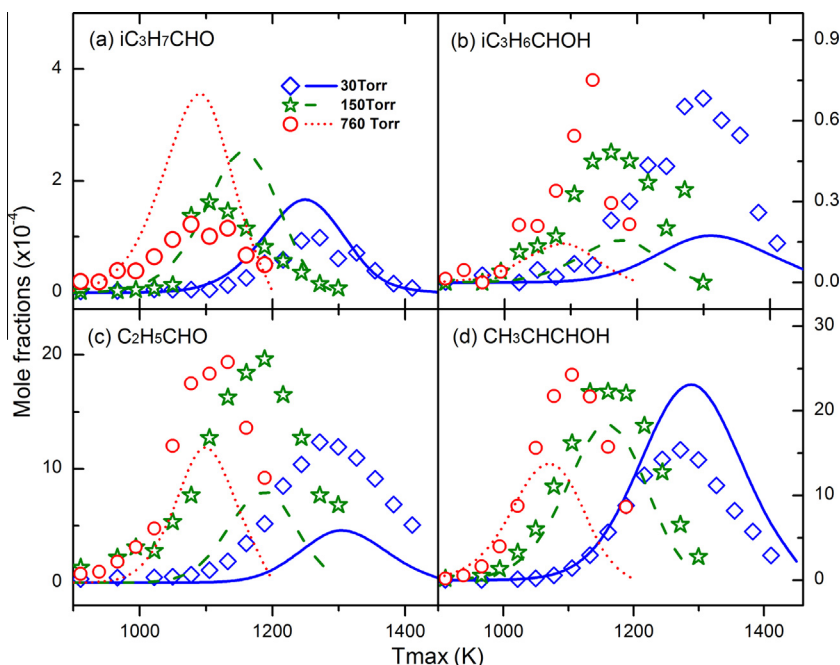
Approximately 10% of  $CH_3CHCHOH$  reacts with H producing  $C_2H_3OH$ , which is the dominant pathway of  $C_2H_3OH$  formation.  $C_2H_3OH$  is not stable; it easily converts to  $CH_3CHO$  through the tautomerization reaction.  $C_2H_3OH$  and  $CH_3CHO$  were detected in this work and their mole fractions are presented in Fig. 7 with simulated results. Figure 7 also shows the experimental and simulated results of  $CH_2O$  and CO.  $CH_2OH$  is the most important precursor of  $CH_2O$ . The  $CH_2OH$  is mainly formed from the C1-C2 scission of  $i$ -butanol (R1) as well as the  $\beta$ -C-C scission of  $iC_4H_8OH-3$  (R12).  $CH_2O$  is mainly consumed by two-step H-abstraction reactions to produce CO.

High concentrations of 1,3- $C_4H_6$  were observed in the experiment, as shown in Fig. 8, whose maximum mole fractions can reach to  $1 \times 10^{-3}$ . The combination reactions of two smaller intermediates such as  $C_2H_3 + C_2H_3$  or  $C_2H_3 + C_2H_4$  cannot explain the experimental observation. A possible route suggested by Schenk et al. [45] is via a  $C_5H_{10}$  (2-methyl-1-butene) intermediate, which is primarily formed from the reaction between  $iC_4H_7$  radical and  $CH_3$  radical, the subsequent decomposition of  $C_5H_9$  radical is likely to form  $C_4H_6 + CH_3$ . This route was tested in the present model; however, it did not work to the formation of  $C_4H_6$ . Zhang et al. proposed a reaction sequence of  $iC_4H_8 \rightarrow iC_4H_7 \rightarrow CH_2 = CHCH^+CH_3 \rightarrow 1,3-C_4H_6$  to explain the high concentration of 1,3- $C_4H_6$  detected in  $iC_4H_8$  pyrolysis [19]. These pathways are included in the present model. As shown in Fig. 8, the mole fraction of 1,3- $C_4H_6$  is well predicted with including the reaction pathways proposed by Zhang et al. However, detailed theoretical study on this sequence is desired to prove the assumption. Figure 8 also presents the mole fractions of  $C_4H_4$ . ROP analysis shows that  $C_4H_4$  is mainly produced by two-step H-abstraction reactions of 1,3- $C_4H_6$ .

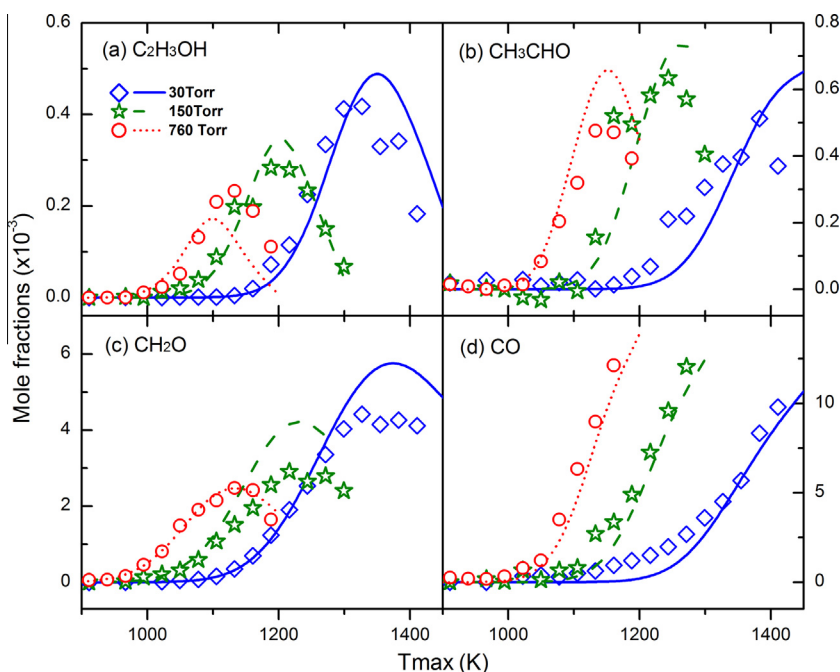
The H-abstraction reactions of  $i$ -butanol lead to the formation of  $iC_4H_8OH-3$  radical also. Most of  $iC_4H_8OH-3$  radical decomposes through  $\beta$ -C-C scission producing  $C_3H_6$  and  $CH_2OH$  (R12). Besides R12, the reaction sequence  $iC_4H_9OH \rightarrow iC_3H_7 \rightarrow C_3H_6$  is another important source of  $C_3H_6$ . The results of C3 hydrocarbons are



**Fig. 5.** Sensitivity analysis of  $iC_4H_8$  from  $i$ -butanol pyrolysis in laminar flow reactor at  $T_{max} = 1450$  K under 30 Torr (black) and  $T_{max} = 1100$  K under 760 Torr (gray). Only those reactions with sensitivities greater than 0.03 are listed.



**Fig. 6.** Experimental mole fraction profiles (symbols) and modeling results (lines) of  $iC_3H_7CHO$ ,  $iC_3H_6CHOH$ ,  $C_2H_5CHO$  and  $CH_3CHCHOH$  from *i*-butanol pyrolysis in a laminar flow reactor at various pressures.



**Fig. 7.** Experimental mole fraction profiles (symbols) and modeling results (lines) of  $C_2H_3OH$ ,  $CH_3CHO$ ,  $CH_2O$  and  $CO$  from *i*-butanol pyrolysis in a laminar flow reactor at various pressures.

illustrated in Fig. 9.  $aC_3H_4$  and  $pC_3H_4$  are mainly produced by reaction sequence  $iC_4H_9OH + H/OH/CH_3 \rightarrow iC_4H_8OH-2 \rightarrow iC_4H_8 \rightarrow iC_4H_7 \rightarrow aC_3H_4 \rightarrow pC_3H_4$ .

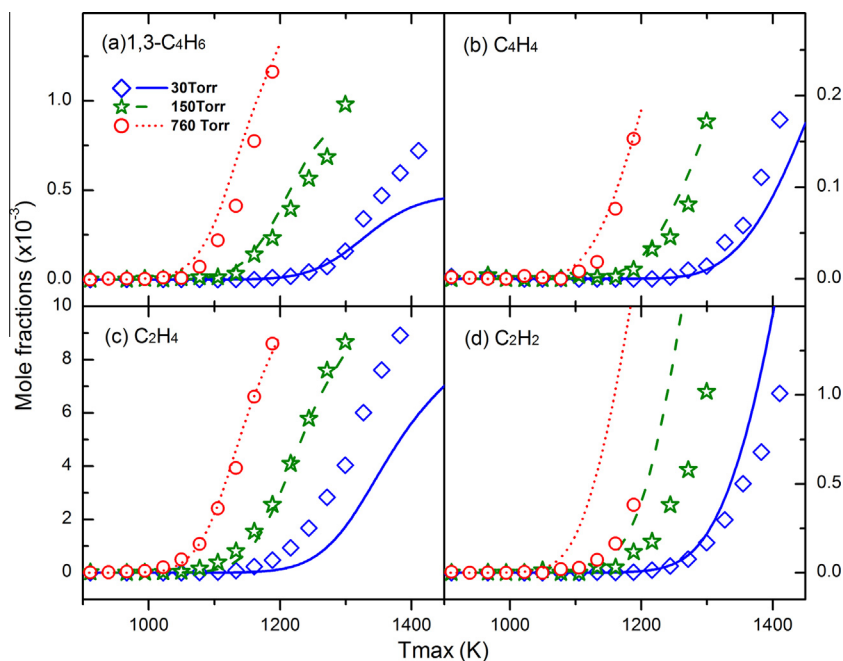
Based on the ROP analysis, the main reaction pathways of *i*-butanol pyrolysis are illustrated in Fig. 10. Since the reaction pathways of *i*-butanol pyrolysis are similar in 30 Torr and 760 Torr, only the reaction network at 760 Torr is illustrated. Most of the stable species in Fig. 10 have been detected by the experiment. It is concluded that the hydrocarbon species are mainly produced through the further decomposition of  $iC_4H_8$  which is produced by

$H_2O$  elimination reaction of *i*-butanol and the reaction sequence of  $iC_4H_9OH + H/OH/CH_3 \rightarrow iC_4H_8OH-2 \rightarrow iC_4H_8$ , while the oxygenated species are mainly produced by the reaction sequence of  $iC_4H_9OH + H/OH/CH_3 \rightarrow iC_4H_8OH-1 \rightarrow iC_3H_7CHO/CH_3CHCHOH$  and its further dissociation pathways.

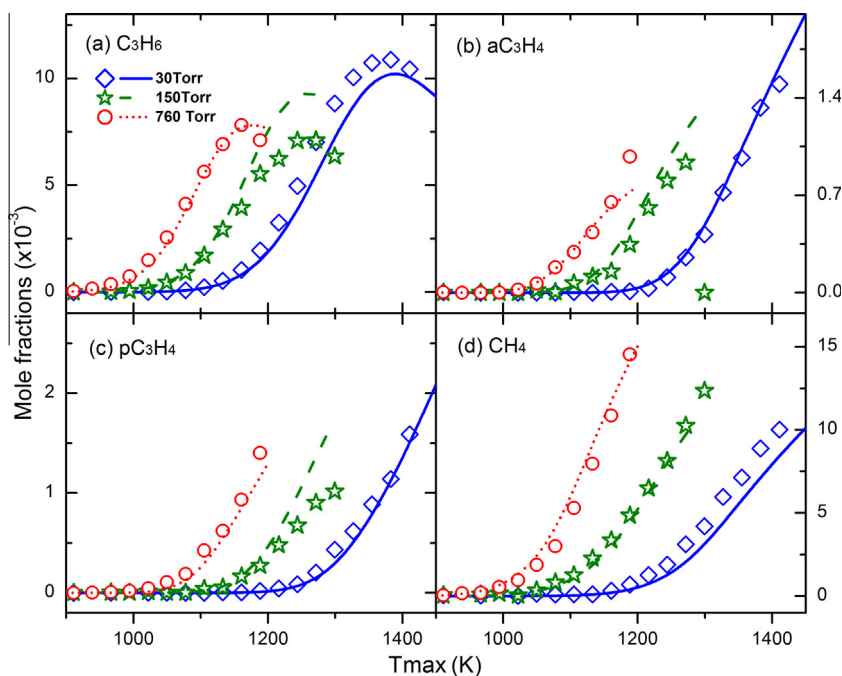
## 5.2. *i*-Butanol pyrolysis in shock tube

The pyrolysis of an *i*-butanol/argon mixture behind reflected shock waves was investigated recently by Stranic et al. [6]. The





**Fig. 8.** Experimental mole fraction profiles (symbols) and modeling results (lines) of 1,3-C<sub>4</sub>H<sub>6</sub>, C<sub>4</sub>H<sub>4</sub>, C<sub>2</sub>H<sub>4</sub> and C<sub>2</sub>H<sub>2</sub> from *i*-butanol pyrolysis in a laminar flow reactor at various pressures.

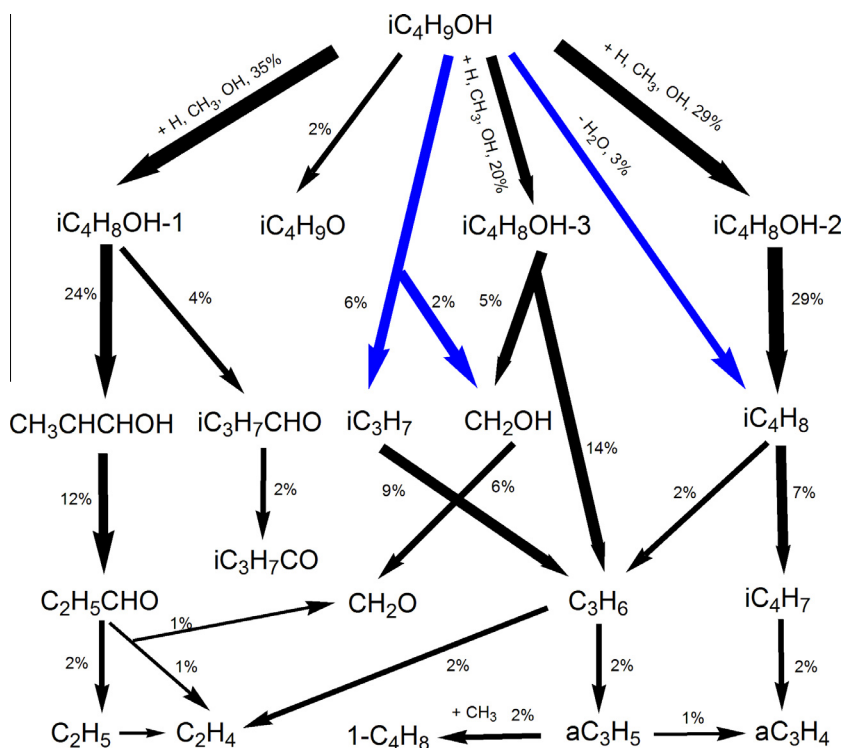


**Fig. 9.** Experimental mole fraction profiles (symbols) and modeling results (lines) of C<sub>3</sub>H<sub>6</sub>, aC<sub>3</sub>H<sub>4</sub>, pC<sub>3</sub>H<sub>4</sub> and CH<sub>4</sub> from *i*-butanol pyrolysis in a laminar flow reactor at various pressures.

initial reflected shock temperatures and pressures cover 1270–1640 K and 1.3–1.9 atm; and time history measurements of species OH, H<sub>2</sub>O, C<sub>2</sub>H<sub>4</sub>, CH<sub>4</sub>, and CO were acquired by using UV and IR laser absorption. Although several kinetic models for *i*-butanol combustion were available including Van Geem et al. model [46], Grana et al. model [15] and Sarathy et al. model [14], Stranic et al. compared their experimental data [6] with simulated results with Sarathy et al. model [14]. Unfortunately, the model cannot predict the species time history satisfactorily. Thus Stranic et al. modified some reactions in Sarathy et al. model according to the results of

sensitivity analysis to get a relatively better prediction. However, there are still discrepancies between experimental and simulated results, which are beyond the experimental uncertainties especially on the prediction of OH mole fractions.

The experimental results of Stranic et al. [6] were used to validate the present model. The simulations were conducted in Chemkin-Pro using the adiabatic Constant-Volume (CV) model by solving energy equations. ROP and sensitivity analyses were also performed to further investigate the pyrolysis behavior of *i*-butanol. The analyses were all conducted at 1477 K and 1.77 atm since

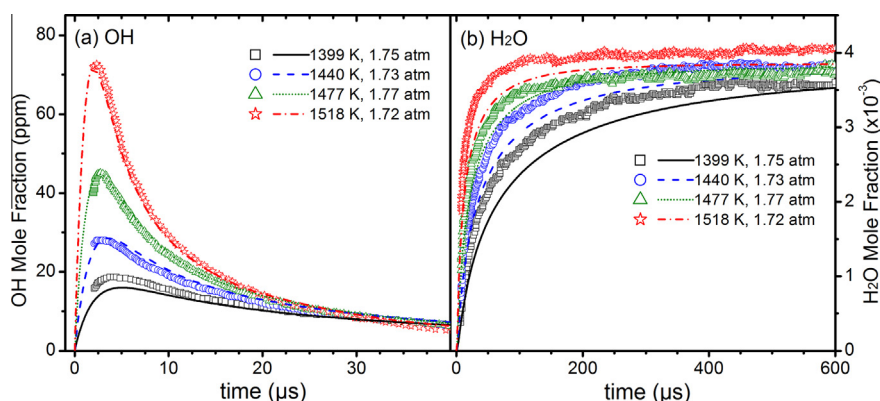


**Fig. 10.** Reaction network of  $iC_4H_9OH$  decomposition at  $T_{max} = 1100$  K and 760 Torr. Thickness of each arrow represents the carbon flux of corresponding reaction(s). The percentage in the figure means the carbon flux of the pathway divided by total carbon flux of  $iC_4H_9OH$  consumption. The blue arrows are the unimolecular pathways.

the main reaction pathways and sensitive reactions are almost identical at all experimental conditions.

The experimental and simulated results of OH and  $H_2O$  time history during *i*-butanol pyrolysis behind shock waves are presented in Fig. 11. The uncertainty of measured OH and  $H_2O$  is  $\pm 5\%$  [6,47]. The mole fraction of OH reaches the peak value at approximately 2  $\mu s$ , which is much earlier than the other detected species. ROP analysis shows that the initial step of *i*-butanol pyrolysis in the shock tube is C1–C2 scission producing  $CH_2OH$  and  $iC_3H_7$ . The dissociation of  $CH_2OH$  and  $iC_3H_7$  produces large amount of H atoms at early times. Then OH radical is mainly produced by reaction sequence of  $iC_4H_9OH + H \rightarrow iC_4H_8OH-2 \rightarrow OH$ . The sensitivity analysis in Fig. 12 shows that four reactions including R1, R13, R15 and R17 have greater sensitivity than the other reactions. OH is mainly consumed by reacting with  $iC_4H_9OH$ . Thus  $iC_4H_9OH + OH$  producing  $iC_4H_8OH-1$  (R15) and  $iC_4H_8OH-3$  (R17) have great

negative sensitivity to OH, as shown in Fig. 12. The reaction R16 shows no sensitivity to OH because this reaction consumes one OH while producing another OH during the decomposition of  $iC_4H_8OH-2$ . Pang et al. [44] measured the  $iC_4H_9OH + OH$  rate constants in shock-heated mixtures of tert-butylhydroperoxide (TBHP) with *i*-butanol excessively diluted in argon. It is found that the rate constants from Hansen et al. [10] show the best agreement with the measurement of Pang et al. [44]. The rate constants of  $iC_4H_9OH + OH$  reactions from Hansen et al. [10] are adopted by the present model, and these reaction rates are considered to be of small uncertainty because they have been validated by the shock tube measurements [44]. R1 and R13 have great positive sensitivity to the formation of OH. Furthermore, R1 is more important at early times ( $<10 \mu s$ ), while R13 is important for the full time history. As shown in Fig. 13, the simulated results match the measurement by using the measured  $k_1$  by Rosado-Reyes and Tsang [42].



**Fig. 11.** Mole fraction profiles of OH and  $H_2O$  from *i*-butanol pyrolysis in the shock tube measured by Stranic et al. [6] (symbols) and predicted results by the present model (lines).

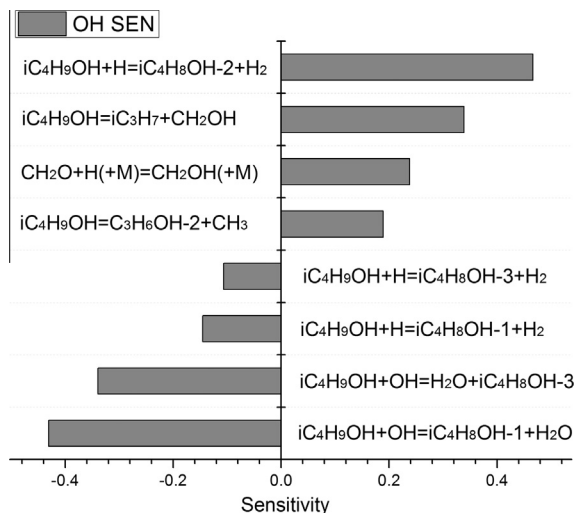
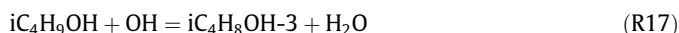
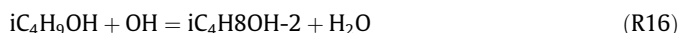
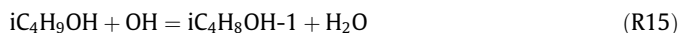
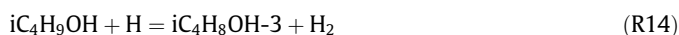
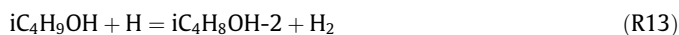


Fig. 12. Sensitivity analysis of OH in *i*-butanol shock tube pyrolysis at 2  $\mu$ s. Initial conditions:  $T = 1477$  K,  $P = 1.77$  atm.



ROP analysis shows that  $H_2O$  mainly originates from the  $H_2O$  elimination reaction of *i*-butanol (R3, 21%) and the reactions between OH and *i*-butanol (45%). The sensitivity analysis of  $H_2O$  is similar to that of OH. Both the H atom formation at early times and the formation of OH radical have great sensitivity to the  $H_2O$  formation.

The predicted and experimental data for the formation of CO and  $CH_4$  behind shock waves are shown in Fig. 14. The uncertainty of the measured CO mole fraction is estimated to  $\pm 10\%$ . It can be seen that the present model can predict the time history of CO within experimental uncertainties at 1485 K, but over-predicts

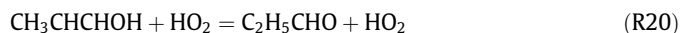
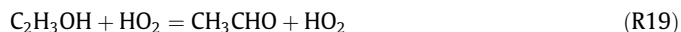
the 1622 K profile by 8% and under-predicts the 1346 K profile by 18%. Most of CO is produced from the reaction sequence of  $iC_4H_9OH \rightarrow CH_2OH \rightarrow CH_2O \rightarrow CHO \rightarrow CO$ , which is similar to the pyrolysis in flow tube. Stranic et al. estimated the uncertainty of the  $CH_4$  mole fraction to be  $\pm 15\%$  [6,47]. However, the  $CH_4$  mole fractions are under-predicted by the present model by appropriate 30% at 1347 K and 1474 K. As proposed by Stranic et al., during the pyrolysis of *i*-butanol, the formation and consumption of  $CH_4$  exhibiting sensitivity to reactions are not directly related to *i*-butanol chemistry. Thus the discrepancy could be caused by both measurement and kinetic model.

### 5.3. Laminar premixed and counter flow non-premixed flames of *i*-butanol

The present model is further validated against a rich laminar premixed *i*-butanol/ $O_2$ /Ar flame at 30 Torr measured by Oßwald et al. [9]. The simulation was performed using the Laminar Premixed Flame Code in the Chemkin-Pro software. The original temperature profile proposed by Oßwald et al. [9] leads to poor prediction, thus the modified one was used to simulate the Oßwald et al. flames by Sarathy et al. [14].

Figure 15 shows the experimental and simulated results of major species in the rich *i*-butanol flame. The present model predicts the mole fractions of inert gas (Ar) and major products ( $H_2$ ,  $H_2O$ , CO) quite well, but it over-predicts the reactivity of *i*-butanol and under-predicts the formation of  $CO_2$ . The mole fraction profiles of selected C1–C4 flame intermediates are presented in Fig. 16. Generally, the present model reproduces most of the experimental results. The reaction network of the rich *i*-butanol flame is illustrated in Fig. 17. The reaction network in the premixed rich flame is quite similar to that in the pyrolysis shown in Fig. 10. Therefore, the following discussions will focus on the difference between pyrolysis and flame.

A ROP analysis shows that *i*-butanol is consumed by unimolecular reactions (12%) and H-abstraction reactions (88%) in the flame. In the flame, 55% of *i*-butanol decomposes through H-abstraction reactions by H atom, and the most preferred products are  $iC_4H_8OH-1$  and  $iC_4H_8OH-2$ .  $iC_4H_8OH-1$  further decomposes mainly via two channels, producing  $CH_3CHCHOH$  and  $iC_3H_7CHO$ .  $CH_3CHCHOH$  and  $iC_3H_7CHO$  are the follow-up products of  $iC_4H_8OH-1$  in both pyrolysis and flame of *i*-butanol. Under the pyrolysis condition, the mole fractions of  $CH_3CHCHOH$  are approximately 20 times higher than those of  $iC_3H_7CHO$  as mentioned above (see Fig. 6). However, in the *i*-butanol flame, the measured mole fractions of  $iC_3H_7CHO$  are higher than those of  $CH_3CHCHOH$  (Fig. 16). In the pyrolysis,  $iC_4H_8OH-1$  is dominantly deconstructed by  $\beta$ -scission producing  $CH_3CHCHOH$  (90%) and  $iC_3H_7CHO$  (10%). However, besides  $\beta$ -scission reactions,  $iC_4H_8OH-1$  is also consumed by reaction with  $O_2$  followed by scission of the O–H bond (R18, 20% of  $iC_4H_8OH-1$  consumption). As a result, the formation channels of  $iC_3H_7CHO$  from  $iC_4H_8OH-1$  in the flame are more competitive than those in the pyrolysis.



Besides  $iC_3H_7CHO$ , three kinds of aldehydes including  $CH_2O$ ,  $CH_3CHO$  and  $C_2H_5CHO$  were detected by Oßwald et al. [9]. ROP analysis shows that  $C_2H_5CHO$  is mainly produced by  $iC_4H_8OH-1 \rightarrow CH_3CHCHOH \rightarrow C_2H_5CHO$ , and the pathway  $iC_4H_8OH-1 \rightarrow CH_3CHCHOH \rightarrow C_2H_5OH \rightarrow CH_3CHO$  is the main pathway for the  $CH_3CHO$  formation. During pyrolysis, enols rapidly isomerize into aldehydes through unimolecular reactions and the H atom assisted

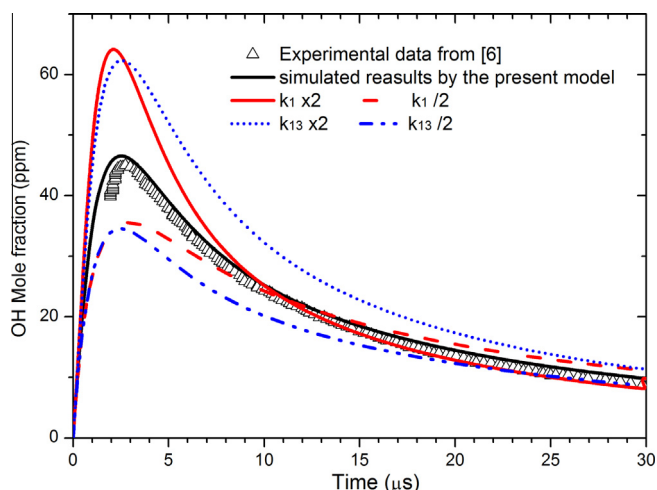
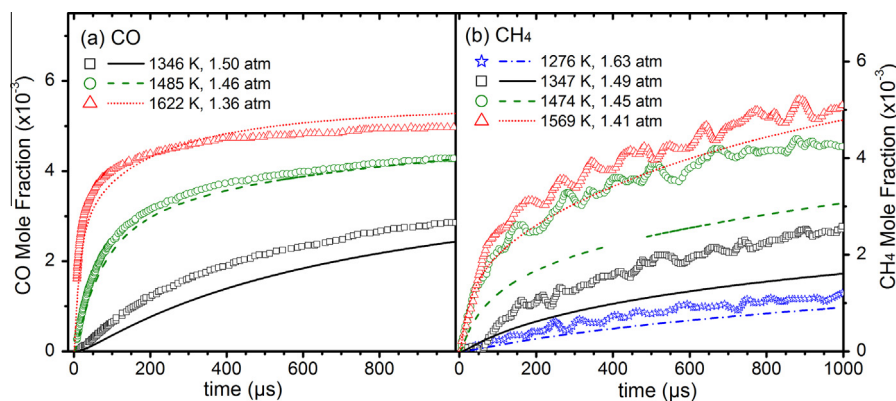
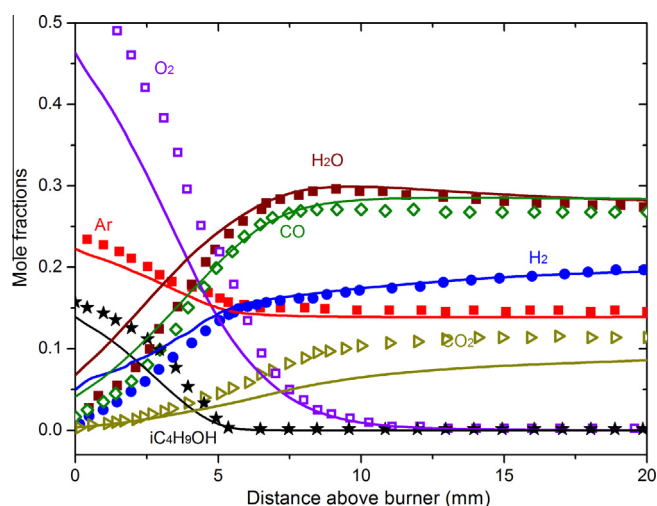


Fig. 13. Mole fraction profile of OH in the shock tube pyrolysis of  $iC_4H_9OH$  at 1477 K and 1.77 atm measured by Stranic et al. [6] (symbols), and the simulation results (lines) by using various rate coefficients of  $iC_4H_9OH = iC_3H_7 + CH_2OH$  (R1) and  $iC_4H_9OH + H = iC_4H_8OH-2 + H_2$  (R13).



**Fig. 14.** Mole fraction profiles of CO and CH<sub>4</sub> from *i*-butanol pyrolysis in a shock tube measured by Stranic et al. [6] (symbols) and predicted results by the present model (lines).



**Fig. 15.** Mole fraction profiles of major species in the premixed rich *i*-butanol flame at 30 Torr measured by Oßwald et al. [9] (symbols) and predicted results by the present model (lines).

tautomerization reactions; while in the flame, the HO<sub>2</sub> assisted tautomerization reactions also play important role in enol consumption (R19, R20). It is concluded that the oxygenated species including iC<sub>3</sub>H<sub>7</sub>CHO, CH<sub>3</sub>CHCHOH, C<sub>2</sub>H<sub>5</sub>CHO, C<sub>2</sub>H<sub>3</sub>OH, CH<sub>3</sub>CHO are mostly produced by the follow-up reactions of iC<sub>4</sub>H<sub>8</sub>OH-1. The reaction pathways of the other hydrocarbon species are similar to those in the pyrolysis, thus are not discussed in detail here.

Grana et al. [15] studied the counter flow non-premixed flame of *i*-butanol using gas chromatography. The experiment was performed at 1 atm with a strain rate of 100 s<sup>-1</sup>, the temperatures of fuel side and air side were 353 K and 298 K, respectively. The detected flame species included reactants, major products and a number of C1–C4 hydrocarbon species. In this work, the concentration profiles of the *i*-butanol non-premixed flame were also simulated to validate the kinetic model. The simulation was performed using the OPDIFF code in Chemkin-Pro software, thermal diffusion (soret) effect was considered and mixture-averaged transport formulism was adopted. The simulation results, compared with the experimental results, are shown in Fig. 18. The present model predicts most of major products well, while it over-predicts the formation of some hydrocarbon species such as *i*-butene, propene, allene and propyne. In the laminar premixed flame, the formation of C<sub>4</sub>H<sub>6</sub> is under-predicted, while in the non-premixed flame, it is over-predicted. As shown in Fig. 18, the analysis shows that the

formation routes and consumption routes of C<sub>4</sub>H<sub>6</sub> in these flames are almost identical. Both the transport properties and the kinetic model may be attributed to the discrepancy between the two flames, and it demonstrates again that more investigation regarding C<sub>4</sub>H<sub>6</sub> chemistry is necessary, especially on its formation routes from *iso*-butene.

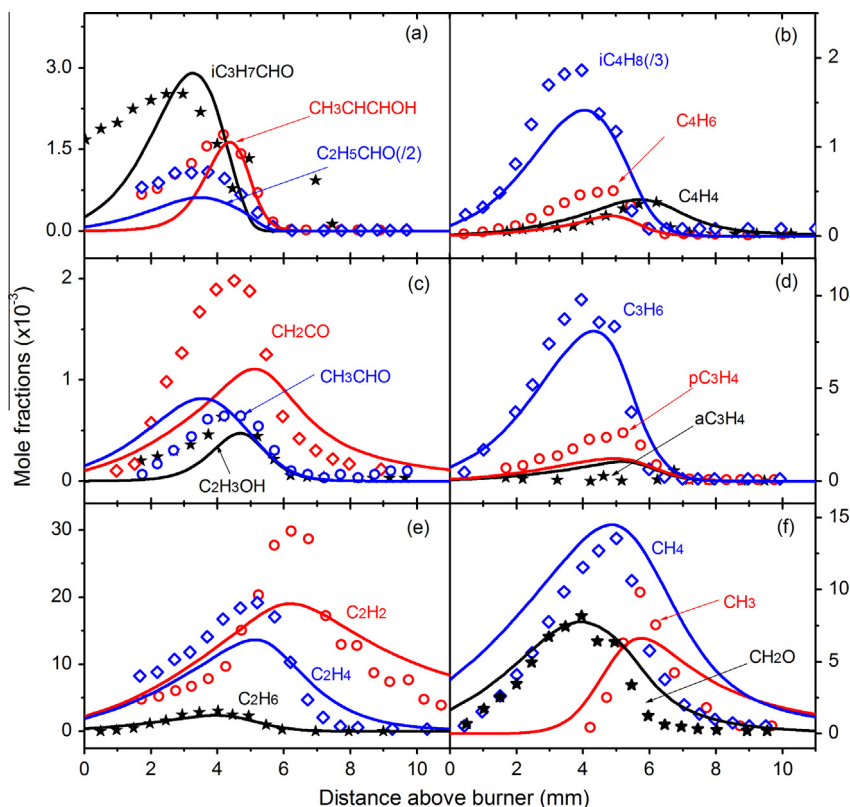
#### 5.4. *i*-Butanol oxidation in jet-stirred reactor

The experiments on the oxidation of *i*-butanol in a jet-stirred reactor at 10 atm over a wide range of equivalence ratios (0.5–4.0) have been performed by Togbé et al. [7]. They used the gas chromatography and infrared spectroscopy to measure the concentration profiles of reactants, stable intermediates and products over the temperature range of 770–1250 K. These data were used for further validation of the present model. The residence time in the experiment is 0.7 s. The simulation was performed using the Perfectly Stirred Reactor code in the Chemkin-Pro program with an end time of 100 s to obtain the steady-state mole fractions of oxidation species.

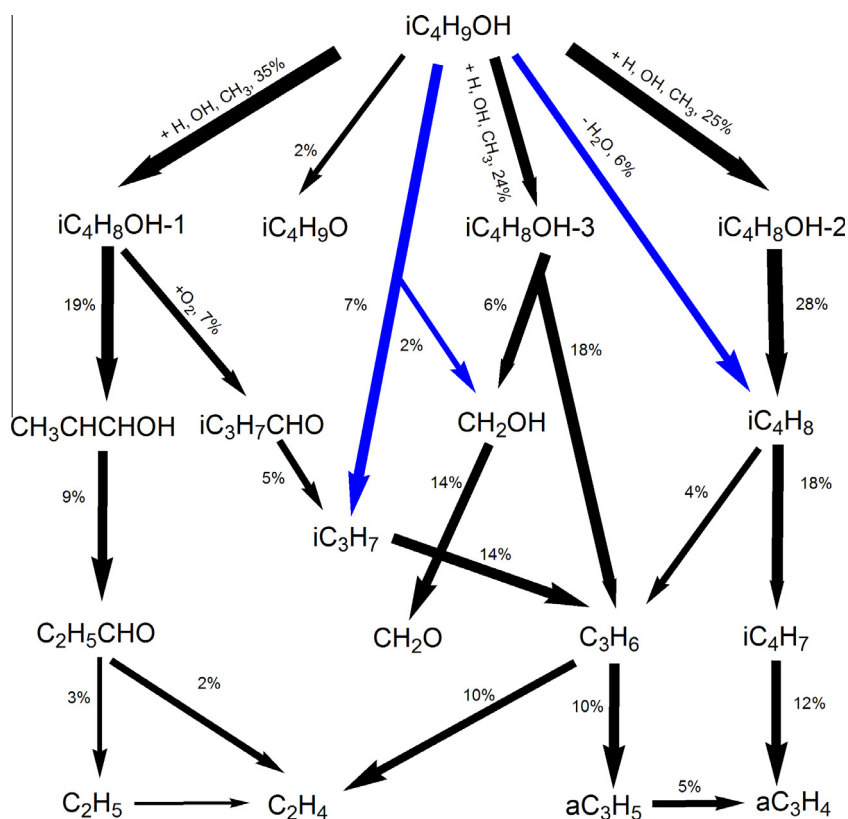
The experimental and simulated results of *i*-butanol oxidation are presented in Figs. 19–21. The discrepancy between the measured and predicted mole fractions of most species is within a factor of 2. Some species have two peaks especially with  $\phi = 4.0$ , such as iC<sub>4</sub>H<sub>8</sub>, H<sub>2</sub>O and CO. This double-peak phenomenon was also observed in 2-butanol oxidation in JSR [7]. The ROP analysis was performed at 800 K and 1000 K to reveal this phenomenon. The ROP analysis at 800 K shows that more than 99% of *i*-butanol decomposes through H-abstraction reactions by small radicals such as H (12%), OH (77%) and HO<sub>2</sub> (7%), as shown in Fig. 22. Among the three H-abstraction products, iC<sub>4</sub>H<sub>8</sub>OH-1 radical is the most dominant one, which contributes to 47% of *i*-butanol consumption. Most of iC<sub>4</sub>H<sub>8</sub>OH-1 decomposes by reacting with O<sub>2</sub>, producing iC<sub>3</sub>H<sub>7</sub>CHO and large amounts of HO<sub>2</sub> radical (R18), which will further decompose into OH. More than 75% of OH is produced from HO<sub>2</sub>. The decomposition temperature of *i*-butanol is about 100 K higher than the experimental results without considering R18; and the first peaks of iC<sub>4</sub>H<sub>8</sub>, H<sub>2</sub>O, CO and C<sub>2</sub>H<sub>4</sub> then disappear over the temperature range of 800–900 K. From the kinetic studies of *n*-butanol [22], 2-butanol [39] and *i*-butanol in JSR oxidation as well as the premixed flame, it is found that the  $\alpha$ -radicals produced by the H-abstraction reactions of alcohols readily react with O<sub>2</sub>, producing HO<sub>2</sub> and aldehyde/ketone. These reactions are very important in the alcohol oxidation chemistry, especially at low to intermediate temperatures.

The reaction pathways at 1000 K are quite different from those at 800 K. The ROP analysis at 1000 K shows that iC<sub>4</sub>H<sub>8</sub>OH-1 is

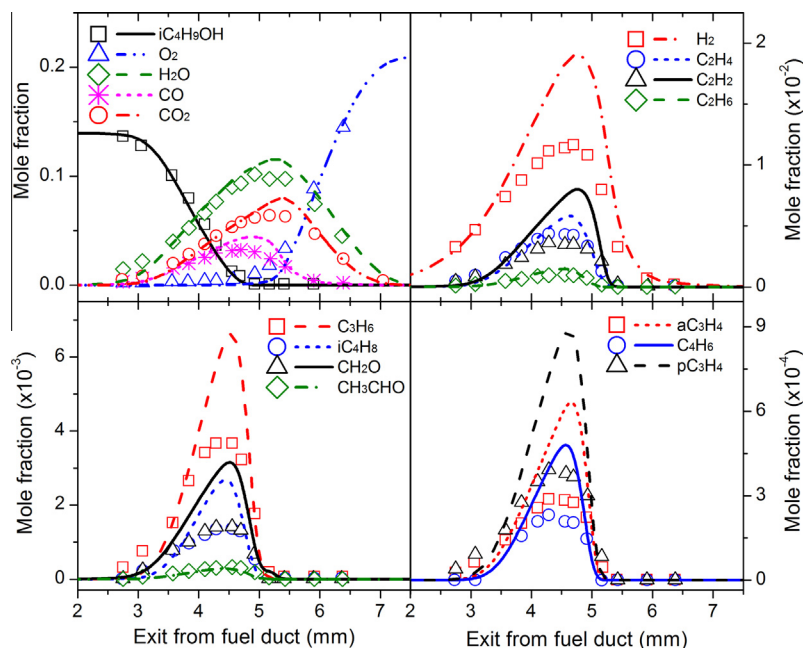




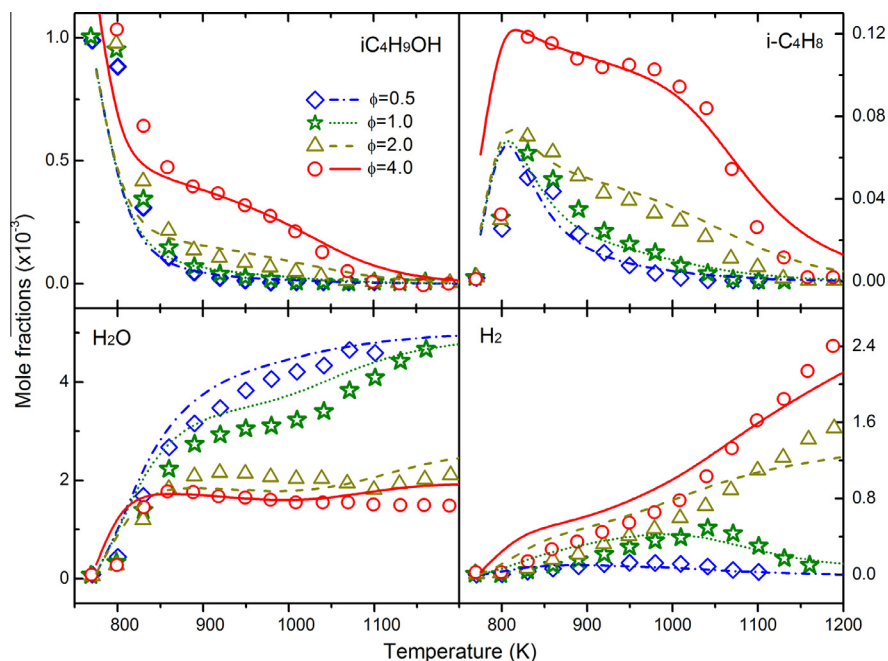
**Fig. 16.** Mole fraction profiles of C1–C4 intermediates in the premixed rich *i*-butanol flame at 30 Torr measured by Oßwald et al. [9] (symbols) and predicted results by the present model (lines).



**Fig. 17.** Reaction network of the premixed rich *i*-butanol flame at 30 Torr. Thickness of each arrow represents the carbon flux of corresponding reaction(s). The percentage in the figure means the carbon flux of the pathway divided by total carbon flux of *i*-butanol consumption. The blue arrows are the unimolecular pathways.



**Fig. 18.** Mole fraction profiles of reactants and products from *i*-butanol counter flow non-premixed flame measured by Grana et al. [15] (symbols) and predicted results by the present model (lines).



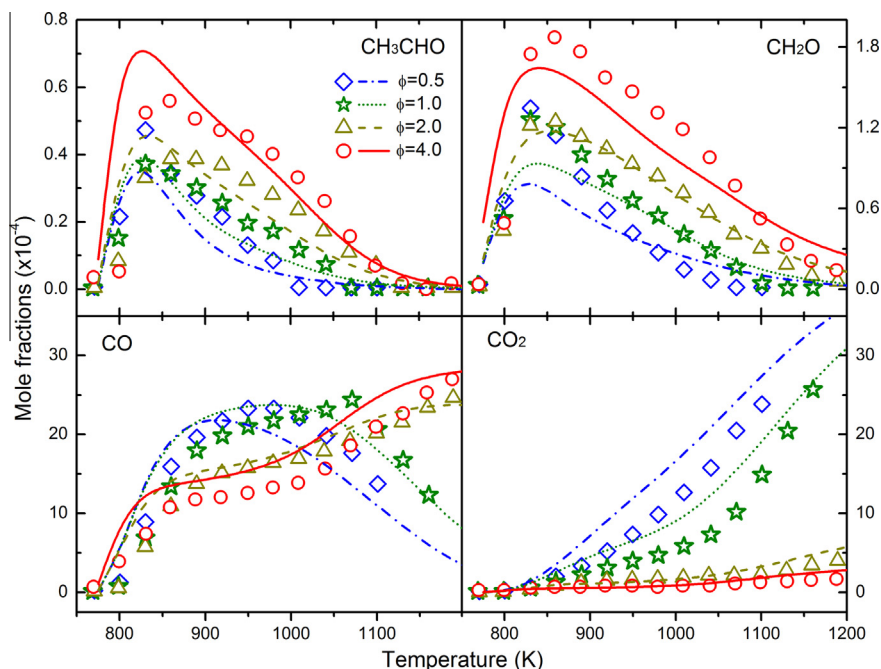
**Fig. 19.** Mole fraction profiles of *i*C<sub>4</sub>H<sub>9</sub>OH, *i*-C<sub>4</sub>H<sub>8</sub>, H<sub>2</sub>O and H<sub>2</sub> from *i*-butanol oxidation in a jet-stirred reactor at 10 atm with four equivalence ratio measured by Togbé et al. [7] (symbols) and predicted results by the present model (lines).

dominated by  $\beta$ -scission reactions, which is the most distinguished feature between 800 K and 1000 K, as shown in Fig. 22. The HO<sub>2</sub> radical mainly originates from the reaction between O<sub>2</sub> and small radicals such as H, HCO, CH<sub>2</sub>OH and C<sub>2</sub>H<sub>5</sub>.

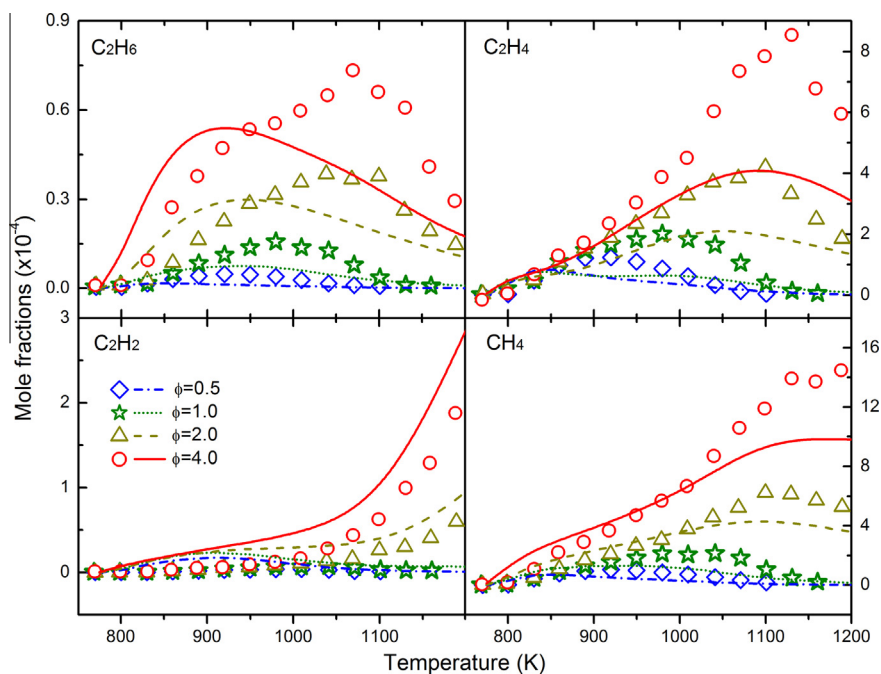
### 5.5. Ignition delay times and flame speeds

The ignition delay times of *i*-butanol behind reflected shock tube have been measured by Moss et al. [12], Stranic et al. [11]

and Yasunaga et al. [5]. Recently, the autoignition delays of *i*-butanol were measured in a heated rapid compression machine (RCM) by Weber et al. [48] at 800–950 K. Yasunaga et al. [5] measured the ignition delay times of *i*-butanol with equivalence ratios varying from 0.5 to 2.0 at 3.5 atm. Stranic et al. [11] also measured *i*-butanol ignition delay times over the wide range of pressures (1.5–45 atm). The present model was validated against the shock tube ignition measurements by running the Chemkin-Pro program, shock tube is assumed as a zero-dimensional homogeneous



**Fig. 20.** Mole fraction profiles of  $\text{CH}_3\text{CHO}$ ,  $\text{CH}_2\text{O}$ ,  $\text{CO}$  and  $\text{CO}_2$  from *i*-butanol oxidation in a jet-stirred reactor at 10 atm with four equivalence ratios measured by Togbé et al. [7] (symbols) and predicted results by the present model (lines).



**Fig. 21.** Mole fraction profiles of  $\text{C}_2\text{H}_6$ ,  $\text{C}_2\text{H}_4$ ,  $\text{C}_2\text{H}_2$  and  $\text{CH}_4$  from *i*-butanol oxidation in a jet-stirred reactor at 10 atm with four equivalence ratios measured by Togbé et al. [7] (symbols) and predicted results by the present model (lines).

adiabatic batch reactor with constant volume. The simulated ignition delay times were defined as the time when the OH concentration reaches half of its maximum. As shown in Fig. 23, most of the data measured by Yasunaga et al. and Stranic et al. are well predicted by the present model. However, the ignition delay times measured by Yasunaga et al. are under-predicted by a factor of 2 at  $\phi = 2.0$ . It is noteworthy that the low temperature oxidation reactions, such as the sub-mechanisms of hydroxyalkyl peroxy radicals, are not included in the present model, thus we cannot get a good prediction on the ignition delay times at lower

temperatures and high pressures performed in RCM by Weber et al. Further development on low temperature oxidation mechanism is required.

Furthermore, Veloo and Egolfopoulos [17] as well as Liu et al. [16] have measured the laminar flame speeds of mixtures of air and *i*-butanol at 1–2 atm. Recently, Wu and Law [13] measured the flame speeds of mixtures of *i*-butanol/ $\text{O}_2/\text{He}$  at 5 atm. These data were also used to validate the present model; the results are illustrated in Fig. 24. The simulation of the flame speeds is performed using the Premix code in Chemkin-Pro software, over

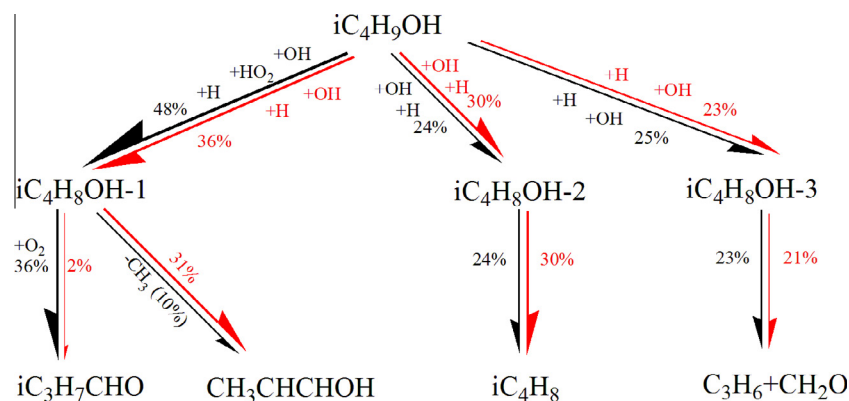


Fig. 22. Primary reaction pathways of  $iC_4H_9OH$  decomposition during JSR oxidation at  $T = 800$  K (black arrow) and  $1000$  K (red arrow). The percentage in the figure denotes the carbon flux of the pathway divided by total carbon flux of  $iC_4H_9OH$  consumption.

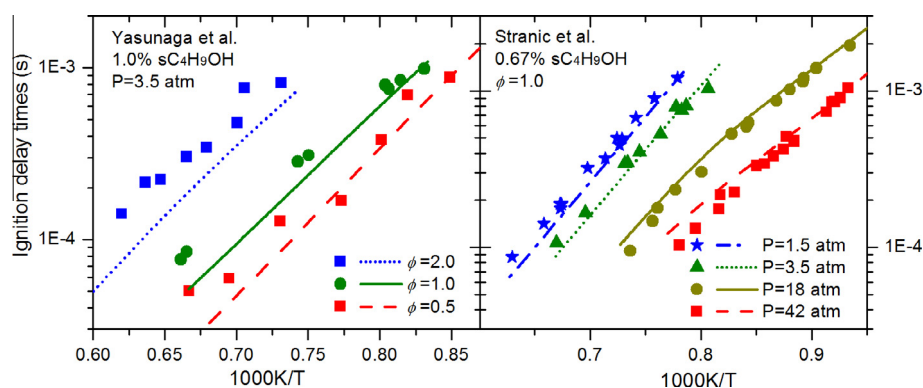


Fig. 23. Ignition delay times of  $i$ -butanol measured by Yasunaga et al. [5] and Stranic et al. [11] (symbols) and predicted results by the present model (lines).

a computational domain of  $15$  cm. The thermal diffusion (soret) effect was considered and mixture-averaged transport formalism was used, the maximum adaptive points were set to  $500$  with high resolution of grid ( $GRAD = 0.01$ ,  $CURV = 0.01$ ), to ensure enough refining of the grid and get the grid-independent results. The uncertainty of the flame speeds measured by Veloo and Egolfopoulos [17] is appropriate  $1.5$  cm/s. However, the flame speeds are slightly under-predicted for  $2.0$  cm/s by the present model.

## 6. Conclusions

The pyrolysis of  $i$ -butanol in the flow reactor at pressures of  $30$ ,  $150$  and  $760$  Torr were studied using SVUV/PIMS over the temperature range of  $900$ – $1550$  K. About  $30$  pyrolysis species were identified in the experiment. Their mole fraction profiles versus the pyrolysis temperatures were determined. A detailed kinetic model consisting of  $186$  species and  $1294$  reactions was developed to simulate the pyrolysis experiments. Generally, the present model predicts most of the pyrolysis species within the experimental uncertainty. Rate constants of some important reactions were updated with recent experimental results and theoretical calculations. The C1–C2 bond scission, producing  $iC_3H_7$  and  $CH_2OH$  radicals, dominates the unimolecular decomposition of  $i$ -butanol. The rate constant ( $k_1$ ) of  $iC_4H_9OH = iC_3H_7 + CH_2OH$  was validated through comparison with three experiments including pyrolysis in both a flow reactor and shock tube, and a premixed flame. ROP and sensitivity analyses were performed to analyze the chemistry of  $i$ -butanol pyrolysis. It is concluded that  $iC_4H_8$  is produced by  $H_2O$  elimination of  $i$ -butanol, and the reaction sequence of  $iC_4H_9OH + H/OH/CH_3 \rightarrow iC_4H_8OH-2 \rightarrow iC_4H_8$  is the main consumption channel for most of the hydrocarbon product species. At the same time, the dissociation of  $iC_4H_8OH-1$  produces most of the oxygenated species.

Furthermore, the present model was validated by laminar premixed  $i$ -butanol flame under low pressure and oxidation experiments in a jet-stirred reactor with equivalence ratios varying from  $0.5$  to  $4.0$  at  $10$  atm. It is found that the reaction between the  $iC_4H_8OH-1$  radical and  $O_2$  plays an important role in the oxidation of  $i$ -butanol, especially at intermediate temperatures. Previously

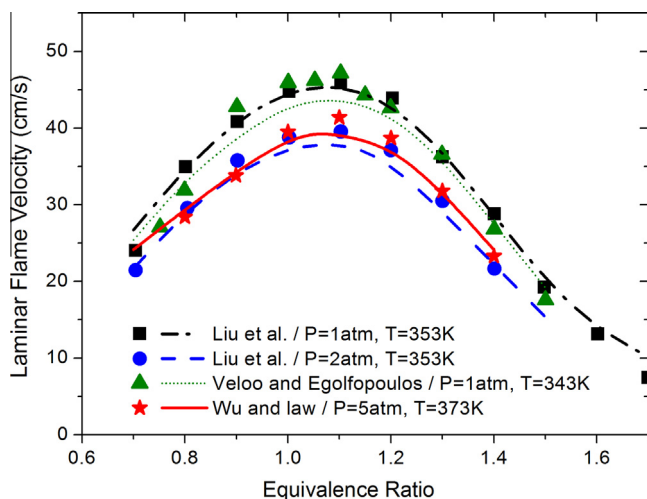


Fig. 24. Flame speed of  $i$ -butanol measured by Veloo and Egolfopoulos [17], Liu et al. [16], and Wu and Law [13] (symbols) and predicted results by the present model (lines).



reported ignition delay times and flame speeds measurements were also used to validate the current model. Good agreement was obtained between simulated and experimental results for pyrolysis in a flow reactor and shock tube, oxidation in a JSR, ignition delay times and laminar flame speeds. It is indicated that the present *i*-butanol model is suitable for wide range conditions with the temperature ranging from 800 to 2200 K, the pressure varying from 5 to 7600 Torr, and the equivalence ratios ranging from 0.25 to pyrolysis conditions.

## Acknowledgments

Authors are grateful to Dr. Enoch Dames for helpful discussion, and the funding supports from National Basic Research Program of China (973 Program) (2013CB834602), Natural Science Foundation of China (51127002 and 21303174), and Chinese Academy of Sciences.

## Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2014.02.004>.

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